MATH 152 **QUIZ 9A** NAME Solutions

Mrs. Bonny Tighe 25 points

11.9 TA_____ Wed 11/15/2021

1. Find the power series representation for the following functions.

a) $f(x) = \dfrac{x}{1+3x^3}$ Find the interval of convergence for this series only.

5pts

$$\frac{x}{1-(-3x^3)} = \sum_{n=0}^{\infty} x(-3x^3)^n = \boxed{\sum_{n=0}^{\infty} (-1)^n 3^n x^{3n+1}}$$

$$|r| = |-3x^3| < 1 \qquad |3x^3| < 1 \qquad |x^3| < 1/3$$

$$-1/3 < x^3 < 1/3$$

$$\sqrt[3]{-1/3} < x < \sqrt[3]{1/3}$$

$$R = \sqrt[3]{1/3}$$

$$I = \left(-\sqrt[3]{1/3}, \sqrt[3]{1/3}\right)$$

Don't need to test end points because it is a geometric series

b) $f(x) = \ln(x+4)$

3pts

$$f'(x) = \frac{1}{x+4} = \frac{1/4}{1-(-x/4)} = \sum_{n=0}^{\infty} \frac{1}{4}\left(-\frac{x}{4}\right)^n =$$

$$f'(x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^n}{4^{n+1}} \qquad \int f'(x)\,dx = f(x)$$

$$\int \sum_{n=0}^{\infty} \frac{(-1)^n x^n}{4^{n+1}}\,dx = C + \sum_{n=0}^{\infty} \frac{(-1)^n x^{n+1}}{4^{n+1}(n+1)}$$

$$(C = \ln 4)$$

3 pts c) $f(x) = \dfrac{1}{(x-1)^2}$ $\int f(x)\,dx = \int (x-1)^{-2}\,dx = \dfrac{-1}{x-1}$

$\dfrac{-1}{x-1} = \dfrac{1}{1-x} = \sum_{n=0}^{\infty} x^n$ $\dfrac{d}{dx}\int f(x)\,dx = f(x)$

$\dfrac{d}{dx}\sum_{n=0}^{\infty} x^n = \boxed{\sum_{n=1}^{\infty} n x^{n-1}}$

2. Find the integral of the power series $\int \sum_{n=0}^{\infty} \dfrac{(-1)^{n+1}(x-2)^{n+2}}{5^n\,(2n)}\,dx$

1 pt $C + \sum_{n=0}^{\infty} \dfrac{(-1)^{n+1}(x-2)^{n+3}}{5^n(2n)(n+3)}$

3. Find the derivative of the given power series

$f(x) = \sum_{k=0}^{\infty} \dfrac{(-1)^k(1\cdot 3\cdot 5\cdots(2k+1))\ x^{2k+1}}{3^{k+1}\,k!}$

1 pt $f'(x) = \sum_{k=0}^{\infty} \dfrac{(-1)^k(1.3.5\cdots(2k+1))(2k+1)\,x^{2k}}{3^{k+1}\,k!}$

$f(x) = \sum_{k=0}^{\infty} \dfrac{(-1)^k(1\cdot 3\cdot 5\cdots(2k+1))\ x^{2k}}{7^k\,k!}$

1 pt $f'(x) = \sum_{k=1}^{\infty} \dfrac{(-1)^k(1.3.5\cdots(2k+1))\,2k\,x^{2k-1}}{7^k\,k!}$

5pts

4. Express the function as the sum of power series by first using partial fractions. Find the interval and radius of convergence. $f(x) = \dfrac{1+x}{2-5x+2x^2}$

$(2-x)(1-2x)$

$$\frac{A}{2-x} + \frac{B}{1-2x} = \frac{1+x}{(2-x)(1-2x)}$$

$$A(1-2x) + B(2-x) = 1+x$$

$x=2$ $\quad A(1-4) = 3 \quad A=-1$

$x = 1/2$ $\quad B(2-1/2) = 1+1/2$

$B(3/2) = 3/2$

$B = 1$

$$\frac{1}{1-2x} - \frac{1}{2-x}$$

$$\frac{1}{1-2x} - \frac{1/2}{1-x/2}$$

$$\sum_{n=0}^{\infty} (2x)^n - \frac{1}{2}(x/2)^n$$

$$\sum_{n=0}^{\infty}\left[2^n x^n - \frac{x^n}{2^{n+1}}\right] = \sum_{n=0}^{\infty} x^n\left[2^n - \frac{1}{2^{n+1}}\right]$$

$|r| = |2x|$ for one

$|2x| < 1$

$|x| < 1/2$

$-1/2 < x < 1/2$

$|r| = \left|\frac{x}{2}\right|$ for other

$\left|\frac{x}{2}\right| < 1$

$|x| < 2$

$-2 < x < 2$

they cross for interval

$I = (-1/2, 1/2)$

$R = 1/2$

5. Rewrite the series $f(x) = \sum_{n=1}^{\infty} \frac{(-1)^n \; x^{2n+1}}{3^{n+1} \; n!}$ here

2pt

$$\sum_{n=0}^{\infty} \frac{(-1)^{n+1} \; x^{2n+3}}{3^{n+2} \; (n+1)!}$$

6. Approximate the integral using power series and 4 non-zero terms.

$$\int_0^{0.2} \frac{x}{4+x^2}\,dx = \int_0^{0.2} \frac{\frac{x}{4}}{1+\frac{x^2}{4}}\,dx = \int_0^{0.2} \sum_{n=0}^{\infty} \frac{x}{4}\left(-\frac{x^2}{4}\right)^n dx$$

4 pts

$$\int_0^{0.2} \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{4^{n+1}}\,dx = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+2}}{4^{n+1}(2n+2)} \Bigg]_0^{0.2}$$

$$\frac{x^2}{4(2)} - \frac{x^4}{4^2(4)} + \frac{x^6}{4^3(6)} - \frac{x^8}{4^4(8)} \Bigg]_0^{0.2}$$

$n=0$ $\quad n=1$ $\quad n=2$ $\quad n=3$

$$\frac{(0.2)^2}{4(2)} - \frac{(0.2)^4}{4^2(4)} + \frac{(0.2)^6}{4^3(6)} - \frac{(0.2)^8}{4^4(8)} - 0$$